

Operating Systems Exam 1 2026/03/25

正式考卷（答案卷）

【114下】535502作業系統 Operating System 第一次考試

姓名：_____ 學號：_____ 班級：_____

考試時段：2026/03/25 10:10 ~ 2026/03/25 12:20 作答限時：130 分鐘

題數：44 滿分：101 分

作答說明

1. ## Post-Exam Instructions
2. After you finish and submit your exam, please complete the following steps:
3. ### 1. Download Your Answering Report
4. Once you have submitted your exam on QJudge, click the **Download** button to save your answering report.
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12. ## Exam Instructions and Rules

Q1. How many times does the following code print "hello"?

```
main() {  
    if (fork() == 0) {  
        fork(); printf("hello\n");  
    };  
    printf("hello\n");  
}
```

(2分, 簡答題)

答案

5 (The child forks again inside the IF; the parent only prints once).

Q2. What is the result of this code?

```
main() {  
    fork();  
    execl("/bin/echo", "echo", "hello", 0);  
    printf("done\n");  
}
```

(2分, 簡答題)

答案

"hello" is printed twice; "done" is never printed. (execl replaces the process image).

Q3. List two advantages of kernel-level threads over user-level threads.

(2分, 問答題)

答案

a. Multiprocessor utilization (can run on different CPUs). b. One thread blocking doesn't block the entire process.

Q4. What is busy waiting (spinlock)?

(2分, 問答題)

答案

Busy waiting, also known as a spin lock, is the situation when a thread is looping, continuously checking whether a condition exists that will allow it to continue. It is often used to check if a thread can enter a critical section or to poll a device to see if data is ready (or written/sent).

Q5. Describe the concept of "demand paging" as a virtual memory implementation. Why is it commonly used?

(2分, 問答題)

答案

a. Demand paging loads pages into memory only when they are needed (on page fault), not at process start. b. It is commonly used because it reduces initial load time, saves memory by not loading unused pages, and allows larger programs than physical memory.

Q6. What is the difference between a preemptive scheduler and a non-preemptive scheduler? Provide an example scenario where preemptive scheduling is beneficial.

(3分, 問答題)

答案

a. Preemptive scheduling can interrupt a running thread before its burst ends to switch to another. b. Non-preemptive runs a thread to completion or voluntary yield. c. Preemptive is beneficial in time-sharing systems for fair CPU sharing among interactive users, preventing long-running tasks from monopolizing the CPU.

Q7. Explain Belady's Anomaly in the context of page replacement algorithms. Which replacement algorithm has this anomaly?

(2分, 問答題)

答案

a. Belady's Anomaly occurs when increasing the number of page frames leads to more page faults. b. The FIFO replacement has Belady's Anomaly. It occurs because FIFO doesn't consider access patterns; older pages might be replaced despite future use.

Q8. What is priority inversion? Describe a scenario where busy waiting (spinlock) can lead to it. How can priority inheritance solve this issue?

(3分, 問答題)

答案

a. Priority inversion occurs when a higher-priority task is blocked by a lower-priority one. b. Scenario: Low-priority A holds a lock; high-priority B spins waiting. The scheduler runs B over A, preventing A from releasing the lock. c. Priority inheritance solves this by temporarily boosting the priority of the lock holder (A) to the level of the waiting high-priority task (B).

Q9. In a system with a TLB hit ratio of 85%, if TLB access time is 15ns and memory access time is 100ns, calculate the effective memory access time (EAT).

(2分, 簡答題)

答案

$EAT = 0.85 * (15 + 100) + 0.15 * (15 + 100 + 100) = 0.85 * 115 + 0.15 * 215 = 97.75 + 32.25 = 130ns.$

Q10. Given a disk with 150 tracks (0-149), head at track 80 moving towards increasing numbers. Requests are in tracks of 20, 45, 10, 120, 130, 5, 90. What is the service order using SCAN and C-SCAN algorithms?

(2分, 問答題)

答案

a. SCAN: 90, 120, 130, 45, 20, 10, 5. (Moves to the end, then reverses). b. C-SCAN: 90, 120, 130, 5, 10, 20, 45. (Moves to the end, jumps to the start, then continues).

Q11. List three attributes that are shared among threads in the same process, and three that are not shared.

(6分, 問答題)

答案

a. Shared: Memory space, file descriptors, global variables. b. Not shared: Thread ID, registers, stack.

Q12. Briefly describe the concepts of "thread," "process," and "program." How do they relate?

(3分, 問答題)

答案

a. A program is a static file containing code. b. A process is an active execution instance of that program. c. A thread is a unit of execution within a process. A process can contain multiple threads that share the same resources.

Q13. In a UNIX inode with 12 direct blocks and 1 single indirect block, assuming a block size of 8KB and address size of 4 bytes, calculate the maximum file size.

(2分, 簡答題)

答案

Total = 16,480 KB (~16.096 MB). Calculation: Direct = $12 * 8\text{KB} = 96\text{KB}$. Single Indirect = $(8\text{KB} / 4\text{B}) * 8\text{KB} = 2048 * 8\text{KB} = 16,384\text{KB}$.

Q14. List the four necessary conditions that must hold simultaneously for a system-wide deadlock to occur.

(4分, 問答題)

答案

Mutual Exclusion; Hold and Wait; No Preemption; Circular Wait.

Q15. Describe the three requirements that a solution to the Critical Section problem must satisfy to be considered correct.

(3分, 問答題)

答案

Mutual Exclusion (only one in CS); Progress (selection of next process cannot be delayed indefinitely); Bounded Waiting (limit on how many times others can enter before a request is granted).

Q16. Explain the difference between 'Internal Fragmentation' and 'External Fragmentation'. Which one is typically associated with the Paging memory management scheme?

(3分, 問答題)

答案

a. Internal Fragmentation: Unused memory within an allocated fixed-size block. b. External Fragmentation: Total free memory exists but is not contiguous. c. Paging is associated with Internal Fragmentation (in the last page).

Q17. Compare the 'Many-to-One' and 'One-to-One' multithreading models. Which one allows better utilization of multi-core processors, and why?

(3分, 問答題)

答案

a. Many-to-One maps many user threads to one kernel thread. b. One-to-One maps each user thread to a kernel thread. c. One-to-One is better for multi-core because it allows multiple threads to run in parallel; Many-to-One can only run one thread at a time because the kernel only sees one entity.

Q18. What is the 'Aging' technique in priority scheduling, and what specific problem does it solve?

(2分, 簡答題)

答案

a. Aging gradually increases the priority of processes that wait in the system for a long time. b. It solves the problem of Starvation (indefinite blocking of low-priority processes).

Q19. What is the primary purpose of the 'Translation Lookaside Buffer' (TLB) in a paging system?

(2分, 簡答題)

答案

To speed up logical-to-physical address translation by caching frequently used page table entries, reducing the need for multiple memory accesses per reference.

Q20. What is “local replacement” and “global replacement”?

(2分, 簡答題)

答案

- a. Local replacement: a process can only select a replacement frame from its own set of allocated frames.
- b. Global replacement: a process may select a replacement frame from the set of all memory frames.

Q21. What are the parameters of Multilevel Feedback Queue Scheduling? (List at least three)

(3分, 問答題)

答案

- a. number of queues
- b. scheduling algorithms for each queue
- c. methods used to determine when to upgrade a process
- d. methods used to determine when to demote a process
- e. methods used to determine which queue a process will enter when that process needs service

Q22. What is the difference between a mode switch and a context switch?

(2分, 簡答題)

答案

- a. Mode switch: change CPU execution mode from one privilege level to another, e.g., user → kernel via a trap or system call.
- b. Context switch: save one process' execution context & restore that of another process.

Q23. A system has 32-bit addresses and 2 GB (2³¹) main memory where each page is 1 MB (2²⁰). How many entries of a page table will contain?

(2分, 簡答題)

答案

Each page is 1 MB (2²⁰), so it needs 20 bits as an offset. Thus a 32-bit address can have at most 12-bit (32-20=12) as the page number. Therefore a page table can have at most $2^{12} = 4096$ entries.

Q24. Please briefly describe the Direct Memory Access (DMA)?

(2分, 問答題)

答案

Direct Memory Access (DMA) is a technique used by computers to transfer data between devices without involving the CPU. DMA allows devices like hard drives, sound cards, and network cards to read or write data directly to or from the main memory, without the need for the CPU to intervene in every transfer, which would waste a significant amount of CPU time. DMA controllers can access the memory independently of the CPU and transfer large amounts of data at high speed, thereby reducing CPU usage and improving system performance.

Q25. What is the race condition?

(2分, 簡答題)

答案

A race condition occurs when two or more threads or processes access a shared resource such as a variable, file, or database record, and the final outcome of the program depends on the order in which the threads or processes execute.

Q26. A major problem with the base & limit approach to memory translation is:

(2分, 單選題)

- ☐ **A.** It requires the process to occupy contiguous memory locations.
- ☐ **B.** The translation process is time-consuming.
- ☐ **C.** The translation must be done for each memory reference.
- ☐ **D.** A process can easily access memory that belongs to another process.

答案

(A) It requires the process to occupy contiguous memory locations.

Q27. Which state transition is not valid?

(2分, 單選題)

- ☐ **A.** Running → Ready
- ☐ **B.** Ready → Blocked
- ☐ **C.** Ready → Running
- ☐ **D.** Running → Blocked

答案

(B) Ready → Blocked

Q28. Which disk scheduling algorithm is most vulnerable to starvation?

(2分, 單選題)

- ☐ **A.** SCAN.
- ☐ **B.** LOOK.
- ☐ **C.** Shortest Seek Time First (SSTF).
- ☐ **D.** First Come, First Served (FCFS)

答案

(C) Shortest Seek Time First (SSTF).

Q29. A semaphore puts a thread to sleep:

(2分, 單選題)

- ☐ A. if it tries to decrement the semaphore's value below 0.
- ☐ B. if it increments the semaphore's value above 0.
- ☐ C. until another thread issues a notify on the semaphore.
- ☐ D. until the semaphore's value reaches a specific number.

答案

(A) if it tries to decrement the semaphore's value below 0.

Q30. There are three processes on a system with the following performance requirements:

. process A runs every 90 milliseconds, using 30 milliseconds of CPU time. . process B runs every 60 milliseconds, using 20 milliseconds of CPU time . process C runs every 500 milliseconds, using 10 milliseconds of CPU time

Use rate-monotonic analysis to assign priorities to the three processes. Assume that priorities range from 0 to 90, with 0 being the highest. Which of the following is a valid set of priority assignments?

(2分, 單選題)

- ☐ A. PA=40, PB=30, PC=100
- ☐ B. PA=40, PB=60, PC=20
- ☐ C. PA=70, PB=50, PC=60
- ☐ D. PA=50, PB=40, PC=90

答案

(D) PA=50, PB=40, PC=90

Q31. Which type of user thread and kernel thread mapping cannot take advantage of a multiprocessor?

(2分, 單選題)

- ☐ A. 1:1
- ☐ B. 1:M
- ☐ C. N:M
- ☐ D. N:1

答案

(D) N:1

Q32. Which of the following statements about mount mechanism and the Virtual File System (VFS) is correct?

(2分, 單選題)

- ☐ **A.** The mount mechanism is used to divide one file into multiple blocks.
- ☐ **B.** The purpose of VFS is to allow a modern operating system to support only one file system.
- ☐ **C.** The mount mechanism can combine multiple file systems into a single hierarchical namespace; VFS provides an abstraction layer that allows different file systems to be supported transparently.
- ☐ **D.** VFS exists only in network devices, not in file systems.

答案

(C) The mount mechanism can combine multiple file systems into a single hierarchical namespace; VFS provides an abstraction layer that allows different file systems to be supported transparently.

Q33. What is the main purpose of a timer interrupt in an operating system?

(2分, 單選題)

- ☐ **A.** To allow user programs to execute privileged instructions directly.
- ☐ **B.** To periodically return control to the OS so it can preempt processes, perform context switching, poll hardware status, and do bookkeeping.
- ☐ **C.** To copy data between disk and memory without CPU involvement.
- ☐ **D.** To make a blocked process skip the ready queue.

答案

(B) To periodically return control to the OS so it can preempt processes, perform context switching, poll hardware status, and do bookkeeping.

Q34. Which of the following statements about demand paging is correct?

(2分, 單選題)

- ☐ **A.** A valid bit indicates that the page is not in memory.
- ☐ **B.** An invalid bit can only indicate an out-of-bound reference.
- ☐ **C.** If there is an invalid reference, the OS will bring the missing page into memory; if the page is simply not in memory, the process will be aborted.
- ☐ **D.** A valid bit means the page is in memory; an invalid bit may indicate either an illegal reference or that the page is not in memory.

答案

(D) A valid bit means the page is in memory; an invalid bit may indicate either an illegal reference or that the page is not in memory.

Q35. Thrashing occurs when:

(2分, 單選題)

- ☐ **A.** The sum of working sets of all processes exceeds available memory.
- ☐ **B.** The TLB hit ratio drops below 50%.
- ☐ **C.** Processes are switched too infrequently.
- ☐ **D.** I/O bursts overlap completely.

答案

(A) The sum of working sets of all processes exceeds available memory.

Q36. Which scheduling criterion measures the time from submission to completion?

(2分, 單選題)

- ☐ **A.** CPU utilization.
- ☐ **B.** Throughput.
- ☐ **C.** Turnaround time.
- ☐ **D.** Waiting time.

答案

(C) Turnaround time.

Q37. The File Allocation Table (FAT) in file systems is a variation of:

(2分, 單選題)

- ☐ **A.** Contiguous allocation.
- ☐ **B.** Combined indexing.
- ☐ **C.** Indexed allocation.
- ☐ **D.** Linked allocation.

答案

(D) Linked allocation.

Q38. Large page sizes in paging systems increase:

(2分, 單選題)

- ☐ **A.** Internal fragmentation.
- ☐ **B.** Page table size.
- ☐ **C.** External fragmentation.
- ☐ **D.** TLB efficiency.

答案

(A) Internal fragmentation.

Q39. Page fault frequency monitoring is used for:

(2分, 單選題)

- ☐ **A.** Optimizing TLB size.
- ☐ **B.** Reducing I/O bursts.
- ☐ **C.** Managing resident sets.
- ☐ **D.** Increasing priority inversion.

答案

(C) Managing resident sets.

Q40. Which CPU scheduling algorithm is mathematically proven to provide the minimum average waiting time for a given set of processes?

(2分, 單選題)

- ☐ **A.** FCFS
- ☐ **B.** Round Robin
- ☐ **C.** Priority Scheduling
- ☐ **D.** Shortest Job First (SJF)

答案

(D) Shortest Job First (SJF)

Q41. Which deadlock prevention strategy works by requiring a process to request all its required resources at once?

(2分, 單選題)

- ☐ **A.** Mutual Exclusion
- ☐ **B.** Hold and Wait
- ☐ **C.** No Preemption
- ☐ **D.** Circular Wait

答案

(B) Hold and Wait

Q42. 'Thrashing' is characterized by which of the following system behaviors?

(2分, 單選題)

- ☐ **A.** Very high paging activity and low CPU utilization.
- ☐ **B.** High CPU utilization.
- ☐ **C.** Rapid file deletions.
- ☐ **D.** Excessive disk capacity.

答案

(A) Very high paging activity and low CPU utilization.

Q43. Peterson's Solution is a classic software-based solution for the Critical Section problem designed for:

(2分, 單選題)

- ☐ **A.** A single process.
- ☐ **B.** Exactly two processes.
- ☐ **C.** Any number of processes.
- ☐ **D.** Real-time systems.

答案

(B) Exactly two processes.

Q44. In an Inverted Page Table, the number of entries in the table is equal to the number of:

(2分, 單選題)

- ☐ **A.** Virtual pages in logical memory.
- ☐ **B.** Threads in the process.
- ☐ **C.** Processes in the system.
- ☐ **D.** Physical frames in physical memory.

答案

(D) Physical frames in physical memory.

答案總表

長答題請參考上方各題詳解。

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Q10	見詳解	2
Q11	見詳解	6
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Q13	見詳解	2
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Q15	見詳解	3
Q16	見詳解	3
Q17	見詳解	3
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題號	答案	配分
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Q21	見詳解	3
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Q26	A	2
Q27	B	2
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Q31	D	2
Q32	C	2
Q33	B	2
Q34	D	2
Q35	A	2
Q36	C	2
Q37	D	2
Q38	A	2
Q39	C	2

題號	答案	配分
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Q41	B	2
Q42	A	2
Q43	B	2
Q44	D	2